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A Survey of Two-Dimensional Automata Theory

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Abstract. The main purpose of this paper is to survey several properties of alternating, non-deterministic, and deterministic two-dimensional Turing machines (including two-dimensional finite automata and marker automata), and to briefly survey cellular types of two-dimensional automata.

1. Introduction

During the past thirty years, many investigations about automata on a one-dimensional tape (i.e., string) have been made (for example, see [25]). On the other hand, since Blum and Hewitt [3] studied two-dimensional finite automata and marker automata, several researchers have been investigating a lot of properties about automata on a two-dimensional tape.

The main purpose of this paper is to survey main results of two-dimensional sequential automata obtained since [3], and to give several open problems. Chapter 2 concerns alternating, nondeterministic, and deterministic two-dimensional Turing machines (including finite automata and marker automata). Section 2.1 gives preliminaries necessary for the subsequent discussions. Section 2.2 gives a difference among alternating, nondeterministic, and deterministic machines. Section 2.3 gives a difference between three-way and four-way machines. Section 2.4 states space complexity results of two-dimensional Turing machines. Sections 2.5 and 2.6 states closure properties and decision problems, respectively. Section 2.7 concerns recognition of connected pictures. Section 2.8 states other topics. Chapter 3 briefly surveys cellular types of two-dimensional automata.

2. Alternating, Nondeterministic, and Deterministic Turing Machines

This chapter concerns alternating, nondeterministic, and deterministic two-dimensional Turing machines, including two-dimensional finite automata and marker automata.

2.1. Preliminaries

Let Σ be a finite set of symbols. A two-dimensional tape over Σ is a two-dimensional rectangular array of elements of Σ . The set of all two-dimensional tapes over Σ is denoted by $\Sigma^{(2)}$.

For a tape $x \in \Sigma^{(2)}$, we let $Q_1(x)$ be the number of rows of x and $Q_2(x)$ be the number of columns of x . If $1 \leq i \leq Q_1(x)$ and $1 \leq j \leq Q_2(x)$, we let $x(i,j)$ denote the symbol in x with coordinates (i,j) . Furthermore, we define

$$x[(i,j),(i',j')]_1$$

when $1 \leq i \leq i' \leq Q_1(x)$ and $1 \leq j \leq j' \leq Q_2(x)$, as the two-dimensional tape z satisfying the following: (i) $Q_1(z) = i' - i + 1$ and $Q_2(z) = j' - j + 1$, (ii) for each k, r [$1 \leq k \leq Q_1(z)$, $1 \leq r \leq Q_2(z)$], $z(k,r) = x(k+i-1, r+j-1)$.

We now give some definitions of two-dimensional alternating Turing machines.

Definition 2.1. A two-dimensional alternating Turing machine (ATM) is a seven-tuple $M = (Q, q_0, U, F, \Sigma, \Gamma, \delta)$, where (1) Q is a finite set of states, (2) $q_0 \in Q$ is the initial state, (3) $U \subseteq Q$ is the set of universal states, (4) $F \subseteq Q$ is the set of accepting states, (5) Σ is a finite input alphabet ($\# \notin \Sigma$ is the boundary symbol), (6) Γ is a finite storage tape alphabet ($B \in \Gamma$ is the blank symbol), and (7) $\delta \subseteq (Q \times (\Sigma \cup \{\#\}) \times \Gamma) \times (Q \times (\Gamma - \{B\}) \times \{\text{left, right, up, down, no move}\} \times \{\text{left, right, no move}\})$ is the next move relation.

A state q in $Q - U$ is said to be existential. As shown in Fig.1, the machine M has a read-only rectangular input tape with boundary symbols $\#$ and one semi-infinite storage tape, initially blank. Of course, M has a finite control, an input head, and a storage tape head. A position is assigned to each cell of the storage tape, as shown in Fig.1. A step of M consists of reading one symbol from each tape, writing a symbol on the storage tape, moving the input and storage heads in specified directions (left, right, up, down, or no move for input head, and left, right, or no move for storage head), and entering a new state, in accordance with the next move relation δ .

A configuration of an ATM $M = (Q, q_0, U, F, \Sigma, \Gamma, \delta)$ is an element of $\Sigma^{(2)} \times (N \cup \{0\})^2 \times S_M$, where $S_M = Q \times (\Gamma - \{B\})^* \times N$, and N denotes the set of all positive integers. The first component x of a configuration $c = (x, (i,j), (q, \alpha, k))$ represents the input to M . The second component (i,j) of c represents the input head position. The third component (q, α, k) of c represents the state of the finite control, nonblank contents of the storage tape, and the storage-head position. If q is the state associated with configuration c , then c is said to be universal (existential, accepting) configuration if q is a universal (existential, accepting) state. The initial configuration of M on input x is $I_M(x) = (x, (1,1), (q_0, \lambda, 1))$, where λ denotes the empty string. We write $c \vdash_M c'$ and say c' is a successor of c if configuration c' follows from configuration c in one step of M , according to the transition rules δ . A computation tree of M is a finite, nonempty labeled tree with the properties,

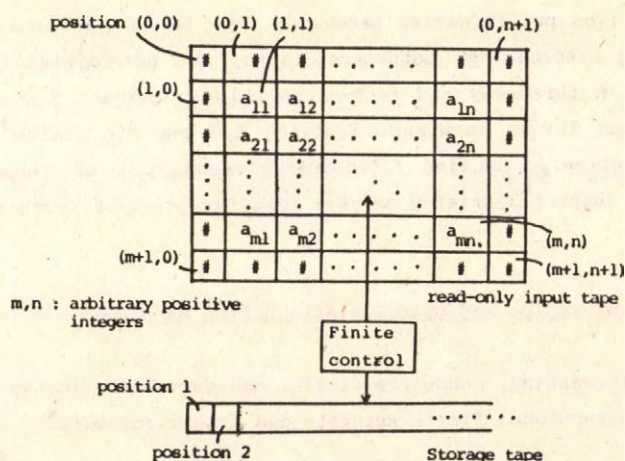


Fig. 1. Two-dimensional alternating Turing machine.

- (1) each node π of the tree is labeled with a configuration $Q(\pi)$,
- (2) if π is an internal node (a nonleaf) of the tree, $Q(\pi)$ is universal, and

$$\{c \mid Q(\pi) \vdash_K c\} = \{c_1, \dots, c_k\},$$
 then π has exactly k children $\rho_1, \dots, \rho_k$ such that $Q(\rho_1) = c_1$,
- (3) if π is an internal node of the tree and $Q(\pi)$ is existential, then π has exactly one child ρ such that $Q(\pi) \vdash_K Q(\rho)$.

An accepting computation tree of M on x is a computation tree whose root is labeled with $I_m(x)$ and whose leaves are all labeled with accepting configurations. We say that M accepts x if there is an accepting computation tree of M on input x . Define $T(M) = \{x \in \Sigma^{(2)} \mid M \text{ accepts } x\}$.

A three-way two-dimensional alternating Turing machine (TATM) is an ATM whose input head can move left, right, or down, but not up.

A two-dimensional nondeterministic Turing machine (NTM) (a three-way two-dimensional nondeterministic Turing machine (TNTM)) is an ATM (TATM) which has no universal state. A two-dimensional deterministic Turing machine (DTM) (a three-way two-dimensional deterministic Turing machine (TDTM)) is an ATM (TATM) whose configurations each have at most one successor.

Let $L(m, n): \mathbb{N}^2 \rightarrow \mathbb{R}$ be a function with two variables m and n , where $\mathbb{R}$ denotes all non-negative real numbers. With each ATM (TATM, NTM, TNTM, DTM, TDTM) M we associate a space complexity function $SPACE$ which takes configuration $c = (x, (i, j), (q, \alpha, k))$ to natural numbers. Let $SPACE(c) = \text{the length of } \alpha$. We say that M is $L(m, n)$ space-bounded if for all $m, n \geq 1$ and for all x with $Q_1(x) = m$ and $Q_2(x) = n$, if x is accepted by M , then there is an accepting computation tree of M on input x such that, for each node π of the tree, $SPACE(Q(\pi)) \leq \lceil L(m, n) \rceil$. By " $ATM(L(m, n))$ ", " $TATM(L(m, n))$ ", " $NTM(L(m, n))$ ", " $TNTM(L(m, n))$ ", " $DTM(L(m, n))$ ", " $TDTM(L(m, n))$ " we denote an $L(m, n)$ space bounded ATM (TATM, NTM, TNTM, DTM, TDTM).

We are also interested in two-dimensional Turing machines M whose input tapes are restricted to square ones. Let $L(m): \mathbb{N} \rightarrow \mathbb{R}$ be a function with one variable m . We say that M is $L(m)$ space-bounded if for all $m \geq 1$ and for all x with $Q_1(x) = Q_2(x) = m$, if x is accepted by M , then there is an accepting computation tree of M on x such that, for each node π of the tree, $SPACE(Q(\pi)) \leq L(m)$. By " $ATM^s(L(m))$ ", " $TATM^s(L(m))$ ", " $NTM^s(L(m))$ ", " $TNTM^s(L(m))$ ", " $DTM^s(L(m))$ ", " $TDTM^s(L(m))$ " we denote an $L(m)$ space-bounded ATM (TATM, NTM, TNTM, DTM, TDTM) whose input tapes are restricted to square ones.

For any constant $k \geq 0$, a k space-bounded ATM (NTM, DTM) is called a two-dimensional alternating (nondeterministic, deterministic) finite automaton, denoted by "AFA" ("NFA", "DFA"). A three-way AFA (NFA, DFA) is denoted by "TAFA" ("TNFA", "TDFA"). For any positive integer k , a two-dimensional alternating (nondeterministic, deterministic) k -marker automaton, denoted by "AMA(k)" ("NMA(k)", "DMA(k)"), is an AFA (NFA, DFA) which can use k markers on the input tape. By "AFA^s" we denote an AFA whose input tapes are restricted to square ones. NFA^s, DFA^s, etc., have the same meaning. Define

$\mathcal{L}[ATM(L(m, n))] = \{T \mid T = T(M) \text{ for some } ATM(L(m, n)) M\}$, and

$\mathcal{L}[ATM^s(L(m))] = \{T \mid T = T(M) \text{ for some } ATM^s(L(m)) M\}$.

$\mathcal{L}[NTM(L(m, n))]$, $\mathcal{L}[NTM^s(L(m))]$, $\mathcal{L}[AFA]$, $\mathcal{L}[AFA^s]$, etc., have the same meaning.

The following concepts are used in the subsequent discussions.

Definition 2.2. A function $L(m): \mathbb{N} \rightarrow \mathbb{R}$ ($L(m, n): \mathbb{N}^2 \rightarrow \mathbb{R}$) is called two-dimensionally space constructible if there is a DTM^s (DTM) M such that (i) for each $m \geq 1$ ($m, n \geq 1$) and for each input tape x with $Q_1(x) = Q_2(x) = m$ ($Q_1(x) = m$ and $Q_2(x) = n$), M uses at most $\lceil L(m) \rceil$ ($\lceil L(m, n) \rceil$) cells of the storage tape, (ii) for each $m \geq 1$ ($m, n \geq 1$), there exists some input tape x with $Q_1(x) = Q_2(x) = m$ ($Q_1(x) = m$ and $Q_2(x) = n$) on which M halts after its storage head has marked off exactly $\lceil L(m) \rceil$ ($\lceil L(m, n) \rceil$) cells of the storage tape, and (iii) for each $m \geq 1$ ($m, n \geq 1$), when given any input tape x with $Q_1(x) =$

$Q_2(x)=m$ ($Q_1(x)=m$ and $Q_2(x)=n$), M never halts without marking off exactly $\lceil L(m) \rceil$ ($\lceil L(m,n) \rceil$) cells of the storage tape.

Definition 2.3. A function $L(m):N \rightarrow R$ ($L(m,n):N^2 \rightarrow R$) is called two-dimensionally fully space constructible if there exists a DTM^{*} (DTM) M which, for each $m \geq 1$ ($m, n \geq 1$) and for each input tape x with $Q_1(x)=Q_2(x)=m$ ($Q_1(x)=m$ and $Q_2(x)=n$), makes use of exactly $\lceil L(m) \rceil$ ($\lceil L(m,n) \rceil$) cells of the storage tape and halts.

Notation 2.1. Let $f(n)$ and $g(n)$ be any functions with one variable n . We write $f(n) \ll g(n)$ when $\lim_{n \rightarrow \infty} f(n)/g(n) = 0$.

2.2. A Difference among Alternating, Nondeterministic, and Deterministic Machines

This section states a difference among the accepting powers of alternating, nondeterministic, and deterministic machines. For the one-dimensional case, it is well known [11,24,69] that the following theorem holds.

Theorem 2.1. For any function $L(n) \ll \log \log n$, $L(n)$ space-bounded two-way alternating, nondeterministic, and deterministic Turing machines are all equivalent to one-way deterministic finite automata in accepting power.

We first show that a different situation occurs for the two-dimensional case. Let $T_1 = \{x \in \{0,1\}^{(2)} \mid \exists m \geq 1 [Q_1(x) = Q_2(x) = m \text{ \& \; } \exists i (1 \leq i \leq m-1) [x[(i,1), (i,m)] = x[(m,1), (m,m)]]]\}$ and $T_2 = \{x \in \{0,1\}^{(2)} \mid \exists m \geq 0 [Q_1(x) = Q_2(x) = 2m+1 \text{ \& \; } x(m+1, m+1) = 1 \text{ (i.e., the center symbol of } x \text{ is 1)}]\}$. It is shown in [58,59] that $T_1 \in \mathcal{L}[\text{TAFAS}] = \mathcal{L}[\text{NTMS}(L(m))]$ and $T_2 \in \mathcal{L}[\text{TNFAS}] = \mathcal{L}[\text{DTMS}(L(m))]$ for any function $L(m) \ll \log m$. Thus we have

Theorem 2.2. For any function $L(m) \ll \log m$, (1) $\mathcal{L}[\text{DTMS}(L(m))] \subsetneq \mathcal{L}[\text{NTMS}(L(m))] \subsetneq \mathcal{L}[\text{ATMS}(L(m))]$, and (2) $\mathcal{L}[\text{TDTM}(L(m))] \subsetneq \mathcal{L}[\text{TNTM}(L(m))] \subsetneq \mathcal{L}[\text{TATM}(L(m))]$.

Corollary 2.1 [3,58,59,89]. $\mathcal{L}[\text{DFA}] \subsetneq \mathcal{L}[\text{NFA}] \subsetneq \mathcal{L}[\text{AFA}]$ and $\mathcal{L}[\text{TDFA}] \subsetneq \mathcal{L}[\text{TNFA}] \subsetneq \mathcal{L}[\text{TAFAS}]$.

For the three-way case, we can show that the following stronger results hold.

Theorem 2.3. (1) $\mathcal{L}[\text{TDTM}(L(m))] \subsetneq \mathcal{L}[\text{TNTM}(L(m))] \subsetneq \mathcal{L}[\text{TATM}(L(m))]$ for any function $L(m) \ll m^2$, (2) $\mathcal{L}[\text{TDTM}(L(m,n))] \subsetneq \mathcal{L}[\text{TNTM}(L(m,n))] \subsetneq \mathcal{L}[\text{TATM}(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$, where $f(m):N \rightarrow R$ is a function such that $f(m) \ll m$, and $g(n):N \rightarrow R$ is a monotone nondecreasing function which is fully space constructible [25], and (3) $\mathcal{L}[\text{TDTM}(L(m,n))] \subsetneq \mathcal{L}[\text{TNTM}(L(m,n))] \subsetneq \mathcal{L}[\text{TATM}(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$, where $f(m):N \rightarrow R$ is a function, and $g(n):N \rightarrow R$ is a function such that $g(n) \ll m$.

Proof. (1): See [44,58].

(2): In [44], it is shown that $\mathcal{L}[\text{TDTM}(L(m,n))] \subsetneq \mathcal{L}[\text{TNTM}(L(m,n))]$. Below, we show that $\mathcal{L}[\text{TNTM}(L(m,n))] \subsetneq \mathcal{L}[\text{TATM}(L(m,n))]$. Let $T[g] = \{x \in \{0,1\}^{(2)} \mid \exists n \geq 1 [Q_1(x) = 2 \times 2^{\lceil g(n) \rceil} \text{ \& \; } Q_2(x) = n \text{ \& \; } (\text{the top and bottom halves of } x \text{ are the same})]\}$. It is easy to show that $T[g] \in \mathcal{L}[\text{TATM}(g(n))]$. The claim follows from this and from the fact [44] that $T[g] \notin \mathcal{L}[\text{TNTM}(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$.

(3): In [44], it is shown that $\mathcal{L}[\text{TDTM}(L(m,n))] \subsetneq \mathcal{L}[\text{TNTM}(L(m,n))]$. Below, we show that $\mathcal{L}[\text{TNTM}(L(m,n))] \subsetneq \mathcal{L}[\text{TATM}(L(m,n))]$. Let $T_3 = \{x \in \{0,1\}^{(2)} \mid Q_1(x) = 2 \text{ \& \; } (\text{the first and second rows of } x \text{ are the same})\}$. It is easy to show that $T_3 \in \mathcal{L}[\text{TAFAS}]$. The claim follows from this and from the fact [44] that $T_3 \notin \mathcal{L}[\text{TNTM}(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$.

For four-way Turing machines on nonsquare tapes, we have

Theorem 2.4. (1) $\mathcal{L}[\text{NTM}(L(m,n))] \subsetneq \mathcal{L}[\text{ATM}(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$, where $f(m):N \rightarrow R$ is a function such that $f(m) \ll \log m$, and $g(n):N \rightarrow R$ is a monotone nondecreasing function which is fully space constructible. (2) $\mathcal{L}[\text{NTM}(L(m,n))] \subsetneq \mathcal{L}[\text{ATM}(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$, where $f(m):N \rightarrow R$ is a monotone nondecreasing function which is fully space con-

structible, and $g(n):N \rightarrow R$ is a function such that $g(n) \ll \log n$.

Proof. We only prove (1), because the proof of (2) is similar. Let $x \in \{0,1\}^{(2)}$ and $Q_2(x)=n$ ($n \geq 1$). When $Q_1(x)$ is divided by $2^{\lceil g(n) \rceil}$, we call

$$x[(j-1)2^{\lceil g(n) \rceil} + 1, 1), (j2^{\lceil g(n) \rceil}, n)]$$

the j -th $g(n)$ -block of x for each j ($1 \leq j \leq Q_1(x)/2^{\lceil g(n) \rceil}$). We say that x has exactly k $g(n)$ -blocks if $Q_2(x)=n$ and $Q_1(x)=k2^{\lceil g(n) \rceil}$ for some positive integer $k \geq 1$. Let $T(g)=\{x \in \{0,1\}^{(2)} \mid (\exists n \geq 1)(\exists k \geq 2)[(x \text{ has exactly } k \text{ } g(n)\text{-blocks}) \& \exists j(2 \leq j \leq k)[\text{the first and } j\text{-th } g(n)\text{-blocks of } x \text{ are identical}]]\}$. It is easy to show that $T(g) \in \mathcal{L}[ATM(g(n))]$. On the other hand, we can show, by using the same technique as in the proof of Lemma 3.3 in [45], that $T(g) \notin \mathcal{L}[NTM(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$. Thus (1) follows.

It is well known [3] that one-dimensional 1-marker automata are equivalent to one-dimensional finite automata. For the two-dimensional case, a different situation occurs. Let T_1 be the set described above. We can show that $T_1 \in \mathcal{L}[DMA(1)] - \mathcal{L}[NFA]$. Let $T_4 = \{x \in \{0,1\}^{(2)} \mid \exists m \geq 1 [Q_1(x)=2m \& Q_2(x)=m \& (\text{the top and bottom halves of } x \text{ are the same})]\}$. It is shown in [29,113] that $T_4 \in \mathcal{L}[NMA(1)] - \mathcal{L}[DMA(1)]$. Thus we have

Theorem 2.5. (1) There exists a set in $\mathcal{L}[DMA(1)]$, but not in $\mathcal{L}[NFA]$, and (2) $\mathcal{L}[DMA(1)] \not\subseteq \mathcal{L}[NMA(1)]$.

Savitch [91] showed that for any fully space constructible function $L(n) \geq \log n$, $L(n)$ space-bounded one-dimensional nondeterministic Turing machines can be simulated by $L^2(n)$ space-bounded one-dimensional deterministic Turing machines. By using the same technique as in [91], we can show that a similar result also holds for the two-dimensional case.

Theorem 2.6. For any two-dimensionally fully space constructible function $L(m) \geq \log m$ ($L(m,n) \geq \log m + \log n$), $\mathcal{L}[NTM^s(L(m))] \subseteq \mathcal{L}[DTM^s(L^2(m))]$ ($\mathcal{L}[NTM(L(m,n))] \subseteq \mathcal{L}[DTM(L^2(m,n))]$).

Open problems: (1) For any two-dimensionally fully space constructible function $L(m) \geq \log m$ ($L(m,n) \geq \log m + \log n$), $\mathcal{L}[DTM^s(L(m))] \subsetneq \mathcal{L}[NTM^s(L(m))] \subsetneq \mathcal{L}[ATM^s(L(m))]$ ($\mathcal{L}[DTM(L(m,n))] \subsetneq \mathcal{L}[NTM(L(m,n))] \subsetneq \mathcal{L}[ATM(L(m,n))]$)? (2) Let $f(m)$ and $g(n)$ be the functions described in Theorem 2.4(1) or Theorem 2.4(2). Then $\mathcal{L}[DTM(L(m,n))] \subsetneq \mathcal{L}[NTM(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$? (3) Is there a set in $\mathcal{L}[NFA]$, but not in $\mathcal{L}[DMA(1)]$? (4) For any $k \geq 1$, $\mathcal{L}[DMA(k)] \subsetneq \mathcal{L}[NMA(k)] \subsetneq \mathcal{L}[AMA(k)]$?

2.3. Three-way versus Four-way

This section states a relationship between the accepting powers of three-way machines and four-way machines.

As shown in Theorem 2.1, for the one-dimensional case, $L(n)$ space-bounded one-way and two-way Turing machines are equivalent for any $L(n) \ll \log \log n$. We shall below show that a different situation occurs for the two-dimensional case. Let $T_5 = \{x \in \{0,1\}^{(2)} \mid \exists m \geq 1 [Q_1(x)=Q_2(x)=2m \& (x[(1,1),(1,m)] \text{ is the reversal of } x[(1,m+1),(1,2m)])]\}$. It is shown in [64] that $T_5 \in \mathcal{L}[DFA^s] - \mathcal{L}[TATM^s(L(m))]$ for any function $L(m) \ll \log m$. On the other hand, as stated in Section 2.2, $T_1 \in \mathcal{L}[TAFAS] - \mathcal{L}[NTM^s(L(m))]$ for any $L(m) \ll \log m$. From these facts, for example, we have

Theorem 2.7. For any function $L(m) \ll \log m$, (1) $\mathcal{L}[TXTM^s(L(m))] \subsetneq \mathcal{L}[XTM^s(L(m))]$ for each $X \in \{D, N, A\}$, (2) $\mathcal{L}[DTM^s(L(m))]$ is incomparable with $\mathcal{L}[TNTM^s(L(m))]$ and $\mathcal{L}[TATM^s(L(m))]$, and (3) $\mathcal{L}[NTM^s(L(m))]$ is incomparable with $\mathcal{L}[TATM^s(L(m))]$.

Remark 2.1. It is shown in [44] that Theorem 2.7(1) can be strengthened as follows: " $\mathcal{L}[TXTM^s(L(m))] \subsetneq \mathcal{L}[XTM^s(L(m))]$ for each $X \in \{D, N\}$ and each function $L(m) \ll m^2$." It is obvious that $\mathcal{L}[TXTM^s(L(m))] = \mathcal{L}[XTM^s(L(m))]$ for each $L(m) \geq m^2$.

Remark 2.2. By using the same technique as in the proof of the fact [74] that $L(n)$ space-bounded

one-way and two-way alternating Turing machines are equivalent for any $L(n) \geq \log n$, we can show that $\mathcal{L}[TATM(L(m))] = \mathcal{L}[ATM(L(m))]$ for any function $L(m) \geq \log m$.

For nonsquare tapes, we have

Theorem 2.8. (1) $\mathcal{L}[XTM(L(m,n))] \subseteq \mathcal{L}[XTM(L(m,n))]$ for each $X \in \{D, N\}$ and each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$, where $f(m)$ and $g(n)$ are the functions described in Theorem 2.3(2) or Theorem 2.3(3), (2) $\mathcal{L}[TATM(L(m,n))] \subseteq \mathcal{L}[ATM(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$, where $f(m): \mathbb{N} \rightarrow \mathbb{R}$ is a function such that $f(m) \ll \log m$, and $g(n): \mathbb{N} \rightarrow \mathbb{R}$ is a monotone nondecreasing function which is fully space constructible, and (3) $\mathcal{L}[TATM(L(m,n))] = \mathcal{L}[ATM(L(m,n))]$ for any function $L(m,n) \geq \log m$.

Proof. See [44] for (1). We leave the proof of (3) to the reader. We below show that (2) holds. Let $T(g)$ be the set described in the proof of Theorem 2.4 (1). As stated in the proof of Theorem 2.4(1), $T(g) \in \mathcal{L}[ATM(g(n))]$. On the other hand, we can show, by using the same technique as in the proof of Lemma 4.2 in [64], that $T(g) \notin \mathcal{L}[TATM(L(m,n))]$ for each $L(m,n) \in \{f(m) \times g(n), f(m)+g(n)\}$. Thus it follows that (2) holds.

It is natural to ask how much space is required for three-way machines to simulate four-way machines. The following two theorems answer this question.

Theorem 2.9. (1) $n \log n (n^2)$ space is necessary and sufficient for TDTM's to simulate DFA's (NFA's) (see [48,83]). (2) n space is necessary and sufficient for TNTM's to simulate DFA's and NFA's (see [57]). (3) $2^{\Theta(n \log n)} (2^{\Theta(n^2)})$ space is necessary and sufficient for TDTM's to simulate DMA(1)'s (NMA(1)'s) (see [67]). (4) $n \log n (n^2)$ space is necessary and sufficient for TNTM's to simulate DMA(1)'s (NMA(1)'s) (see [67]). (In this theorem, note that n denotes the number of columns of tapes.)

Open problems: (1) $\mathcal{L}[AFA] \subseteq \mathcal{L}[TNTM(n)]$? (2) $\mathcal{L}[AMA(1)] \subseteq \mathcal{L}[TNTM(2^{\Theta(n)})]$?

2.4. Two-Dimensionally Space Constructible Functions and Space Complexity Results

This section concerns two-dimensionally space constructible functions and space complexity hierarchy. We state these subjects only for square tapes. (See [78,80,82] for the case of non-square tapes.) It is well known [24] that in the one-dimensional case, there exists no space constructible function which grows more slowly than the order of $\log \log n$, thus no space hierarchy of language acceptability exists below space complexity $\log \log n$. Below, we state that a different situation occurs for the two-dimensional case.

We consider the following three functions:

- (i) $\log^{(1)} m = \begin{cases} 0 & (m=0) \\ \lceil \log_2 m \rceil & (m \geq 1) \end{cases}$
 $\log^{(k+1)} m = \log^{(1)}(\log^{(k)} m)$
- (ii) $\exp^* 0 = 1$, $\exp^*(m+1) = 2^{\exp^* m}$
- (iii) $\log^* m = \min\{x \mid \exp^* x \geq m\}$.

The following theorem demonstrates that there exist two-dimensionally space constructible functions which grow more slowly than the order of $\log \log m$.

Theorem 2.10 [78,82]. The functions $\log^* m$ (k : any natural number) and $\log^* m$ are two-dimensionally space constructible.

More generally, we have

Theorem 2.11 [78,82]. Let $f(m): \mathbb{N} \rightarrow \mathbb{N}$ be any monotone nondecreasing total recursive function such that $\lim_{m \rightarrow \infty} f(m) = \infty$. Then, there exists a two-dimensionally space constructible and monotone non-decreasing function $L(m)$ such that (i) $L(m) < f(m)$ and (ii) $\lim_{m \rightarrow \infty} L(m) = \infty$.

It is shown in [105] that there exists no fully space constructible function which grows more

slowly than the order of $\log m$. It is unknown whether or not there exists a two-dimensionally fully space constructible function which grows more slowly than the order of $\log m$.

For the one-dimensional case, the following three important theorems concerning space complexity hierarchy of Turing machines are known. (By $\mathcal{L}[1NTM(L(n))]$ ($\mathcal{L}[1DTM(L(n))]$) we denote the class of languages accepted by $L(n)$ space-bounded one-dimensional nondeterministic (deterministic) Turing machines [25].)

Theorem 2.12 [102]. Let $L(n)$ be a space function. For any constant $c > 0$ and each $X \in \{D, N\}$, $\mathcal{L}[1XTM(L(n))] = \mathcal{L}[1XTM(c \cdot L(n))]$.

Theorem 2.13 [102]. Let $L_1(n)$ and $L_2(n)$ be any space constructible functions such that $\lim_{i \rightarrow \infty} L_1(n_i)/L_2(n_i) = 0$ and $L_2(n_i)/\log n_i > k$ ($i=1, 2, \dots$) for some increasing sequence of natural numbers $\{n_i\}$ and for some constant $k > 0$. Then there exists a language in $\mathcal{L}[1DTM(L_2(n))]$, but not in $\mathcal{L}[1DTM(L_1(n))]$.

Theorem 2.14 [24]. Let $L_1(n)$ and $L_2(n)$ be space constructible functions such that $\lim_{i \rightarrow \infty} L_1(n_i)/L_2(n_i) = 0$ and $L_2(n_i)/\log n_i < 1/2$ for some increasing sequence of natural numbers $\{n_i\}$. Then there exists a language in $\mathcal{L}[1DTM(L_2(n))]$, but not in $\mathcal{L}[1DTM(L_1(n))]$.

By using the ideas similar to those of the proofs of Theorems 2.12 and Theorem 2.13, we can prove the following two-dimensional analogues to these theorems.

Theorem 2.15. Let $L(m)$ be a space function. For any constant $c > 0$ and each $X \in \{D, N, A\}$,

$$\mathcal{L}[XTM^*(L(m))] = \mathcal{L}[XTM^*(cL(m))].$$

Theorem 2.16 [78, 80]. Let $L_2(m)$ be a two-dimensionally space constructible function. Suppose that $\lim_{i \rightarrow \infty} L_1(m_i)/L_2(m_i) = 0$ and $L_2(m_i) > k \cdot \log m_i$ ($i=1, 2, \dots$) for some increasing sequence of natural numbers $\{m_i\}$ and for some constant $k > 0$. Then there exists a set in $\mathcal{L}[DTM^*(L_2(m))]$ but not in $\mathcal{L}[DTM^*(L_1(m))]$.

Recently, It is shown in [28, 103] that for each space constructible function $L(n) \geq \log n$, $\mathcal{L}[1NTM(L(n))]$ is closed under complementation. This result can be extended to the two-dimensional case. By using these facts, we can extend Theorem 2.13 and Theorem 2.16 to the nondeterministic case [21].

The following theorem, which is a two-dimensional analogue to Theorem 2.14, cannot be proved by the same idea as in the proof of Theorem 2.14.

Theorem 2.17 [78, 80]. Let $L_2(m)$ be a two-dimensionally space constructible function. Suppose that $\lim_{i \rightarrow \infty} L_1(m_i)/L_2(m_i) = 0$, $\lim_{i \rightarrow \infty} L_2(m_i) = \infty$, and $L_2(m_i) < k \cdot \log m_i$ ($i=1, 2, \dots$) for some increasing sequence of natural numbers $\{m_i\}$ and for some constant $k > 0$. Then there exists a set in $\mathcal{L}[DTM^*(L_2(m))]$, but not in $\mathcal{L}[DTM^*(L_1(m))]$.

The following theorem, which is a nondeterministic version of Theorem 2.17, is proved in [60].

Theorem 2.18 [60]. Let $L_2(m)$ be a two-dimensionally space constructible function such that $L_2(m) \leq \log m$. Suppose that $\lim_{i \rightarrow \infty} L_1(m_i)/L_2(m_i) = 0$. Then there exists a set in $\mathcal{L}[NTM^*(L_2(m))]$ (in fact, in $\mathcal{L}[DTM^*(L_2(m))]$) but not in $\mathcal{L}[NTM^*(L_1(m))]$.

From Theorem 2.10 and Theorem 2.18, we have the following corollary, which implies that in the two-dimensional case, there is an infinite hierarchy of acceptabilities even for space complexity classes below $\log \log m$.

Corollary 2.2. For any constant $c > 0$, each $k \in \mathbb{N}$, and each $X \in \{D, N\}$,

$$\mathcal{L}[XFA^*] = \mathcal{L}[XTM^*(c)] \subsetneq \dots \subsetneq \mathcal{L}[XTM^*(\log^{(k+1)} m)] \subsetneq \mathcal{L}[XTM^*(\log^{(k)} m)] \subsetneq \dots$$

Open problem: Do results analogous to Theorems 2.16 and 2.17 hold for ATM^* ?

2.5 Closure properties

This section presents only closure properties of the classes of sets accepted by several types of

two-dimensional finite automata. (See [41,44,45,48,106] for closure properties of the classes of sets accepted by space-bounded two-dimensional Turing machines.) It is well known [25] that the class of sets accepted by one-dimensional finite automata is closed under many operations, including Boolean operations. We below demonstrate that a different situation occurs for two-dimensional finite automata. We first define several operations over two-dimensional tapes.

Definition 2.4. Let

$$x = \begin{matrix} a_{11} \dots a_{1n} \\ \vdots \\ a_{m1} \dots a_{mn} \end{matrix}, \text{ and } y = \begin{matrix} b_{11} \dots b_{1n'} \\ \vdots \\ b_{m'1} \dots b_{m'n'} \end{matrix}$$

Then the rotation x^R of x and the row reflection x^{RR} of x are given by Fig.2 and Fig.3, respectively. A row cyclic shift of x is any two-dimensional tape of the form of Fig.4 for some $1 \leq k \leq m$ (not that for $k=m$ this is x itself), and a column cyclic shift of x is any two-dimensional tape of the form of Fig.5 for some $1 \leq k \leq n$ (not that for $k=n$ this is x itself). The row catenation $x \oplus y$ is defined only when $n=n'$ and is given by Fig.6, and the column catenation $x \odot y$ is defined only when $m=m'$ and is given by Fig.7.

$$\begin{matrix} a_{m1} \dots a_{m1} \\ \vdots \\ a_{m1} \dots a_{m1} \end{matrix}$$

Fig.2

$$\begin{matrix} a_{m1} \dots a_{mn} \\ \vdots \\ a_{11} \dots a_{1n} \end{matrix}$$

Fig.3

$$\begin{matrix} a_{k+1,1} \dots a_{k+1,n} \\ \vdots \\ a_{k+1,1} \dots a_{k+1,n} \end{matrix}$$

Fig.4

$$\begin{matrix} a_{k1} \dots a_{kn} \\ \vdots \\ a_{k1} \dots a_{kn} \end{matrix}$$

Fig.4

$$\begin{matrix} a_{11} \dots a_{1n} \\ \vdots \\ a_{11} \dots a_{1n} \end{matrix}$$

Fig.5

$$\begin{matrix} a_{11} \dots a_{1n} \\ \vdots \\ a_{11} \dots a_{1n} \end{matrix}$$

Fig.5

$$\begin{matrix} a_{1,k+1} \dots a_{1n} & a_{11} \dots a_{1k} \\ \vdots & \vdots \\ a_{m,k+1} \dots a_{mn} & a_{m1} \dots a_{mk} \end{matrix}$$

Fig.6

$$\begin{matrix} a_{11} \dots a_{1n} & b_{11} \dots b_{1n'} \\ \vdots & \vdots \\ a_{m1} \dots a_{mn} & b_{m'1} \dots b_{m'n'} \end{matrix}$$

Fig.7

Definition 2.5. Let S and S' be two sets of two-dimensional tapes. Then

- $S^R = \{x^R \mid x \in S\}$ (rotation of S),
- $S^{RR} = \{x^{RR} \mid x \in S\}$ (row reflection of S),
- $S^{RC} = \{y \mid y \text{ is a row cyclic shift of some } x \in S\}$ (row cyclic closure of S),
- $S^{CC} = \{y \mid y \text{ is a column cyclic shift of some } x \in S\}$ (column cyclic closure of S),
- $S \oplus S' = \{x \oplus y \mid x \in S, y \in S'\}$ (row catenation),
- $S \odot S' = \{x \odot y \mid x \in S, y \in S'\}$ (column catenation),
- $S_+ = \bigcup_{i=1}^{\infty} S_i$ (row closure),
- $S^+ = \bigcup_{i=1}^{\infty} S_i$ (column closure),

where $S_1 = S$, $S_2 = S \oplus S$, ..., $S_{i+1} = S_i \oplus S$, and $S^1 = S$, $S^2 = S \odot S$, ..., $S^{i+1} = S^i \odot S$.

For three-way finite automata, we have

Theorem 2.19. (1) $\mathcal{L}[\text{TDFA}]$ is not closed under union, intersection, rotation, row reflection, row and column cyclic closures, row and column catenations, or row and column closures [44,45,48,56,106]. (2) $\mathcal{L}[\text{TNFA}]$ is closed under union, row catenation, and row closure, but not closed under intersection, complementation, rotation, row and column cyclic closures, column catenation, or column closure [44,45,56,106]. (3) $\mathcal{L}[\text{TAFA}]$ is closed under union and intersection, but not closed under rotation, row reflection, row and column cyclic closures, row and column catenations, or row and column closures [64,68].

Open problems: (1) Are $\mathcal{L}[\text{TDFA}]$ and $\mathcal{L}[\text{TAFA}]$ closed under complementation? (2) Is $\mathcal{L}[\text{TNFA}]$ closed under row reflection?

For four-way finite automata, we have

Theorem 2.20. (1) $\mathcal{L}[\text{DFA}]$ is closed under Boolean operations, rotation and row reflection, but not closed under row and column cyclic closures, row and column catenations, or row and column closures [41,42,51]. (2) $\mathcal{L}[\text{NFA}]$ is closed under union, intersection, rotation, and row reflection, but not closed under row and column cyclic closures, row and column catenations, or row and column closures [41,51,52]. (3) $\mathcal{L}[\text{AFA}]$ is closed under union and intersection, rotation, and row reflection.

Remark 2.3. That $\mathcal{L}[\text{DFA}]$ is closed under Boolean operations can be proved by using the technique in [96].

Open problems: (1) Is $\mathcal{L}[\text{NFA}]$ closed under complementation? (2) Is $\mathcal{L}[\text{AFA}]$ closed under complementation, row and column cyclic closures, row and column catenations, and row and column closures?

2.6. Decision Problems

This section concerns decision problems of two-dimensional finite automata. It is well known [25] that many decision problems of one-dimensional finite automata are decidable. As suggested by the following theorem, most of decision problems of four-way two-dimensional finite automata are undecidable.

Theorem 2.21 [3,111]. The emptiness and universe problems for DFA's are undecidable even for a one-letter alphabet.

We below state some decision problems of three-way finite automata. For each $X \in \{\text{D}, \text{N}, \text{A}\}$, let $\text{TXFA}(0)$ denote a TXFA which operates on two-dimensional tapes over a one-letter alphabet. The following two theorems are all that have been obtained for three-way finite automata by now.

Theorem 2.22 [49]. (1) The emptiness and universe problems for TDFA(0)'s are decidable. (2) The emptiness problem for TNFA(0)'s is decidable. (3) The universe, inclusion, and equivalence problems for TNFA's are undecidable.

Theorem 2.23 [70]. (1) The disjointness, inclusion, and equivalence problems for TDFA(0)'s are decidable. (2) The disjointness and inclusion problems for TDFA's are undecidable.

Open problems: (1) Are the emptiness, universe, and equivalence problems for TDFA's decidable? (2) Are the universe, inclusion, and equivalence problems for TNFA(0)'s and TAFA(0)'s decidable? (3) Is the emptiness problem for TAFA(0)'s decidable?

2.7. Recognizability of Connected Pictures

Let T_c be the set of all two-dimensional connected pictures [53,89]. It is interesting to investigate how much space is required for two-dimensional Turing machines to accept T_c . For this problem, we have

Theorem 2.24. (1) n space is necessary and sufficient for TDTM's and TNTM's to accept T_c (see [116]). (2) $T_c \in \mathcal{L}[\text{AFA}]$ (see [58]). (3) $T_c \in \mathcal{L}[\text{DMA}(1)]$ (see [3,89]). (4) $T_c \notin \mathcal{L}[\text{TATM}^s(L(m))]$ for any $L(m) \ll \log m$, where T_c^s denotes the set of all the square connected pictures (see [64]).

Open problem: $T_c \in \mathcal{L}[\text{DFA}]$ or $T_c \in \mathcal{L}[\text{NFA}]$?

2.8. Other Topics

In this section, we list up other topics and related references about sequential automata on a two-dimensional tape.

(1) Maze (or labyrinth) search problems: see [1,4,5,7,8,9,10,22,75,104].

(2) Characterizations of one-dimensional languages by two-dimensional automata: see [20,29,32,33].

(3) A relationship between two-dimensional automata and two-dimensional array grammars: see [19,30,73,76,79,84,89,99,115].

(4) Properties of special types of two-dimensional Turing machines (two-dimensional pushdown automata, stack automata, multi-counter automata, multihead automata, and marker automata): see [3,27,46,47,55,56,78,81,89,94,95,113].

(5) Parallel, time, space, and reversal complexities of two-dimensional alternating multihead Turing machines: see [26,50,58,59].

(6) Properties of two-dimensional finite automata over a one-letter alphabet: see [36,40,70].

(7) Properties of two-dimensional automata on a nonrectangular tape: see [77,88,89].

(8) A relationship between two-dimensional alternating finite automata and cellular types of two-dimensional automata: see [62,63,65,66].

The most interesting problem in the future is to investigate time complexity hierarchy of two-dimensional Turing machines.

Two-dimensional (or array) grammars are not discussed here. For this subject, see the excellent book of Rosenfeld [89] and the excellent surveys of Siromoney [97,98].

3. Cellular Types of Two-Dimensional Automata

Many authors investigated language acceptability of one-dimensional cellular automata (for example, see [6,12,14,101,114]). On the other hand, cellular automata on a two-dimensional tape are being investigated not only in the viewpoint of formal language theory but also in the viewpoint of pattern recognition. Cellular automata on a two-dimensional tape can be classified into three types.

The first type, called a two-dimensional cellular automaton (CA for short), is investigated in [2,13,17,29,31,34,35,37,39,53,61-63,65,71,72,87,89,100,112]. CA's make use of two-dimensional cellular arrays. It is shown, for example, that (1) the set T_c of all two-dimensional connected pictures can be accepted by deterministic CA's in linear time [2], (2) the majority problem can be solved by deterministic CA's in linear time, and thus the set of all the two-dimensional tapes over $\{0,1\}$ with positive Euler number can be accepted by deterministic CA's in linear time [100], (3) the two-dimensional packing problem can be solved by deterministic CA's in linear time [71], (4) NFA's can be simulated by deterministic CA's in linear time [72], and (5) AFA's can be simulated by deterministic CA's in constant state change [62]. (The notion of state change complexity was first introduced in [114]. Many properties of two-dimensional on-line tessellation acceptors (OTA's for short) introduced in [29,35] are investigated in [29,31,34,35,37,39,53,65,112]. The OTA is a restricted type of CA in which cells do not make transitions at every time step; rather, a transition 'wave' passes once diagonally across the array. It is shown, for example, that (1) non-deterministic OTA's are more powerful than NFA's, and deterministic OTA's are incomparable (in accepting power) with NFA's and DFA's [29,35], (2) the set T_c described above cannot be accepted by deterministic OTA's [53], and (3) deterministic OTA's can be used as two-dimensional pattern matching machines [112]. In [17], a generalization of CA's in which each cell is a space-bounded Turing machine rather than a finite automaton, is introduced. Fast algorithms are given for performing various basic image processing tasks by such automata.

The second type of cellular automata on a two-dimensional tape is investigated in [15,30-33,37,38,57,66,78,89,90,92,93,99,107-110]. Two typical models of this type are parallel/sequential

array automata (PSA's) [90] and one-dimensional bounded cellular acceptors (BCA's) [92,107-110]. The PSA makes use of one-dimensional (e.g., horizontal) cellular array which can move, as a unit, in the vertical direction, and accepts a tape if the leftmost cell (i.e., the cell which reads the first column of the tape) enters an accepting state in some time. The BCA is a restricted type of one-way PSA in which the cellular array moves downwards each time step, and the BCA accepts a tape if the state configuration of the cellular array just after it has completely scanned the tape is an element of the specified regular set (called the accepting configuration set). It is shown, for example, that (1) nondeterministic one-way PSA's are more powerful than deterministic ones, two-way PSA's are more powerful than one-way PSA's, and T_c is accepted by deterministic one-way PSA's [90], (2) deterministic one-way PSA's are incomparable with NFA's and DFA's [15,48], (3) one-way PSA's are more powerful than OTA's [35], (4) nondeterministic BCA's are equivalent to nondeterministic OTA's, and deterministic BCA's are incomparable with deterministic OTA's and DFA's [31,57]. See [30,99] for a relationship between PSA's and two-dimensional grammars, and see [37,38] for closure properties of PSA's. An extension of BCA's in which the accepting configuration set is a context-free language, context-sensitive language, or phrase structure language, is introduced in [107-110].

The third type, called a pyramid cellular acceptor (PCA), is investigated in [16,18,43,54,85,86,89]. The PCA is a pyramid stack of two-dimensional cellular arrays, where the bottom array has size 2^n by 2^n , the next lowest 2^{n-1} by 2^{n-1} , and so forth, the $(n+1)$ st layer consisting of a single cell, called the root. Each cell has nine neighbors -- four son cells in a 2-by-2 block in the level below, four brother cells in the current level, and one father cell in the level above. The transition function of each cell maps 10-tuples of states into states -- or sets of states, in the nondeterministic case. An input tape is stored as initial states of the bottom array; the upper-level cells are initialized to a quiescent state. The root is the accepting state. A bottom-up pyramid cellular acceptor (UPCA) is a PCA in which the next state of a cell depends only on the current states of that cell and its four sons. It is shown, for example, that (1) both nondeterministic PCA's and nondeterministic UPCA's are equivalent to nondeterministic CA's [16,85,89], (2) nondeterministic UPCA's are more powerful than deterministic UPCA's [85,89], (3) nondeterministic UPCA's can simulate nondeterministic OTA's, thus NFA's in $O(\text{diameter})$ time [54,86], and (4) $O(\text{diameter} \times \log \text{diameter})$ time ($O((\text{diameter})^2)$ time) is necessary for deterministic UPCA's to simulate DFA's (NFA's) [54]. See the excellent book [89] of Rosenfeld for image processing task by PCA's and UPCA's.

Open problems:

- (1) Can AFA's be simulated by deterministic CA's in linear time ?
- (2) Can AFA's be simulated by nondeterministic OTA's ?
- (3) Are deterministic CA's equivalent to nondeterministic CA's ?
- (4) Is T_c accepted by nondeterministic OTA's or deterministic UPCA's ?
- (5) Is T_c accepted by nondeterministic UPCA's in diameter time ?
- (6) Can DFA's, NFA's, or AFA's be simulated by deterministic UPCA's ?

4. Conclusions

In this paper, we surveyed several aspects of two-dimensional automata theory. We believe that there are many problems about two-dimensional automata to solve in the future. We hope that this survey will activate the investigation of two-dimensional automata theory.

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